

UK Regional Housing Market Model

Complete Technical Report — Version 0.2.0

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1 Project Overview and Motivation

1.1 Purpose

This report documents the UK Regional Housing Market Model (v0.2.0), a research platform estimating equilibrium fair values and stochastic forward price paths for residential property across 12 UK regions. The model is designed for: (a) systematic research into regional fair value and mean-reversion dynamics; (b) scenario-based risk assessment for long-horizon (≥ 24 month) planning by institutional investors and REIT analysts; and (c) comparative regional analysis anchored to Land Registry HPI data.

The model is *not* suitable for individual property valuation, short-horizon trading decisions, or regulatory capital calculations. It is a historical research prototype and does not constitute regulated financial advice or a stress-testing model as defined under PRA SS3/18.

1.2 Seven-Stage Pipeline

1. **Ingestion** — fetch and cache raw UK housing and macroeconomic data from Land Registry, Bank of England, ONS, and HMRC.
2. **Cleaning** — harmonise, interpolate, and construct the canonical monthly panel; record staleness and coverage metadata.
3. **Pre-OLS checks and feature engineering** — multicollinearity diagnostics and construction of lagged macro features.
4. **OLS calibration** — semi-structural fair-value layer and monthly growth regression; estimation of OU parameters (κ, σ, γ) per region.
5. **Monte Carlo simulation** — 10,000-path Euler-Maruyama simulation per region per scenario over a 5-year horizon.
6. **Scoring engine** — consumer and REIT composite scores derived from simulation distributional statistics.
7. **Streamlit dashboard** — interactive research UI covering all 12 English, Welsh, Scottish, and Northern Irish regions.

1.3 Coverage

- **Regions:** 12 (all English regions, Wales, Scotland, Northern Ireland).
- **Panel start:** 2005-01-01; **effective data as of:** 2025-01-01.
- **Simulation horizon:** 5 years (60 monthly steps), 5 macro scenarios.
- **Total paths:** 600,000 (10,000 paths $\times$ 5 scenarios $\times$ 12 regions).
- **Validation status:** APPROVED — 9/9 sign-off areas complete (2026-04-02).

2 Mathematical Foundations

2.1 Probability Space and Brownian Motion

Definition 2.1. A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is the sample space, $\mathcal{F}$ is a σ -algebra of events, and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure with $\mathbb{P}(\Omega) = 1$.

Definition 2.2. A *standard Brownian motion* (Wiener process) $\{W_t\}_{t \geq 0}$ is a continuous-time stochastic process satisfying:

1. $W_0 = 0$ almost surely;
2. for $0 \leq s < t$, the increment $W_t - W_s \sim \mathcal{N}(0, t - s)$;
3. increments over disjoint intervals are independent;
4. $t \mapsto W_t$ is almost surely continuous.

Key properties used throughout the model:

$$\mathbb{E}[W_t] = 0, \quad \text{Var}(W_t) = t, \quad \mathbb{E}[W_s W_t] = \min(s, t).$$

The quadratic variation identity $\mathbb{E}[(dW_t)^2] = dt$ is the source of the second-order correction in Itô's formula.

2.2 Itô's Lemma

Theorem 2.3 (Itô's Lemma). *Let X_t satisfy the SDE*

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t. \tag{1}$$

Let $f : \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$ be twice continuously differentiable. Then

$$df(X_t, t) = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW_t. \tag{2}$$

Proof. Apply a second-order Taylor expansion in (dx, dt) :

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (dX_t)^2 + \dots$$

Substituting $dX_t = \mu dt + \sigma dW_t$ and applying the Itô multiplication table

$\times$	dt	dW_t
dt	0	0
dW_t	0	dt

gives $(dX_t)^2 = \sigma^2 dt + O(dt^{3/2})$. Collecting terms in dt and dW_t :

$$df = \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dW_t. \quad \square$$

2.3 Geometric Brownian Motion

Geometric Brownian Motion (GBM) is the classical financial price benchmark:

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (3)$$

Proposition 2.4 (GBM Closed-Form Solution). *The explicit solution of the GBM SDE is*

$$S_t = S_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right]. \quad (4)$$

Proof. Apply Theorem 2.3 with $f(S, t) = \log S$. Then $f_t = 0$, $f_S = 1/S$, $f_{SS} = -1/S^2$:

$$d \log S_t = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t.$$

This is a linear SDE with constant coefficients; integrating from 0 to t gives $\log(S_t/S_0) = (\mu - \frac{1}{2}\sigma^2)t + \sigma W_t$, and exponentiating yields the result. $\square$

GBM preserves positivity and is analytically tractable, but it has *no mean reversion*. For housing markets, where valuation gaps attract prices back toward a fundamental anchor, GBM is insufficient as the primary simulation process.

2.4 Log Ornstein-Uhlenbeck Process

Definition 2.5. Let $x_t = \log P_t$ and $m_t = \log P_t^*$. The *log Ornstein-Uhlenbeck* (log-OU) process is

$$dx_t = \kappa (m_t - x_t) dt + \gamma dt + \sigma dW_t, \quad (5)$$

where $\kappa > 0$ is the mean-reversion speed, m_t is the time-varying log fair-value target, γ is the annual drift, and $\sigma > 0$ is annual volatility.

Exact solution for constant m . Rewrite (5) as $dx_t + \kappa x_t dt = (\kappa m + \gamma) dt + \sigma dW_t$. Multiply both sides by the integrating factor $e^{\kappa t}$:

$$d(e^{\kappa t} x_t) = e^{\kappa t} (\kappa m + \gamma) dt + \sigma e^{\kappa t} dW_t.$$

Integrating from 0 to t and dividing by $e^{\kappa t}$:

$$x_t = m + \frac{\gamma}{\kappa} + e^{-\kappa t} \left(x_0 - m - \frac{\gamma}{\kappa} \right) + \sigma \int_0^t e^{-\kappa(t-s)} dW_s. \quad (6)$$

The stochastic integral is Gaussian with zero mean and variance $\sigma^2(1 - e^{-2\kappa t})/(2\kappa)$. The *half-life* of the log-gap is

$$t_{1/2} = \frac{\ln 2}{\kappa}. \quad (7)$$

Remark 2.6. For London ($\kappa = 0.200$), $t_{1/2} \approx 3.47$ years. For Yorkshire and The Humber ($\kappa = 0.326$), $t_{1/2} \approx 2.13$ years. For West Midlands ($\kappa = 0.131$), $t_{1/2} \approx 5.29$ years. Northern Ireland ($\kappa = 0.825$) has the fastest reversion: $t_{1/2} \approx 0.84$ years.

2.5 Euler-Maruyama Discretisation

The model simulates monthly steps using the Euler-Maruyama scheme. Let $\Delta t = 1/12$ and $x_n = x(n \Delta t)$.

$$x_{n+1} = x_n + \kappa(m_{n+1} - x_n) \Delta t + \gamma \Delta t + \sigma \sqrt{\Delta t} \varepsilon_{n+1}, \quad \varepsilon_{n+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (8)$$

The simulated price is recovered by exponentiation: $P_{n+1} = \exp(x_{n+1})$.

Remark 2.7 (Convergence orders). The Euler-Maruyama method has strong convergence order 1/2 and weak convergence order 1. Strong convergence governs pathwise approximation quality; weak convergence governs the accuracy of expected functional quantities. For the distributional statistics (percentiles, probability of loss) reported in Stage 5, weak convergence at order 1 is the relevant criterion.

3 Data and Panel Construction

3.1 Data Sources

Series	Source	Frequency	Coverage
Nominal house prices	HM Land Registry HPI	Monthly	2005–2026
Mortgage rate (2yr fixed)	Bank of England	Monthly	2005–2026
Base rate	Bank of England	Monthly	2005–2026
Gross earnings (EARN01)	ONS	Annual	2005–2025
Private rents	ONS / Valuation Office	Monthly	2010–2026
CPI	ONS	Monthly	2005–2026
Transactions (SDLT)	HMRC	Monthly	2005–2026
Mortgage approvals	Bank of England	Monthly	2005–2026
Net housing additions	DLUHC	Annual	2005–2025

3.2 Panel Construction

The canonical panel is built at monthly regional frequency. Annual series (earnings, net additions) are linearly interpolated to monthly, and the panel carries explicit staleness flags for all interpolated columns. As of the modelling date (2026-04-02), earnings staleness is 13 months across all regions.

Key panel statistics: 2,532 rows $\times$ 41 columns, 12 regions, starting 2005-01-01. The fair-value OLS uses *anchor months* only (June and December of each year), yielding $n = 452$ observations in the pooled regression.

3.3 Feature Engineering

The monthly growth model uses the following constructed features.

- `mortgage_rate_lag3`: Bank of England 2-year fixed mortgage rate, lagged three months.
- `financial_stress_excess_lag3`: excess of a financial-stress composite index above a threshold, lagged three months.
- `earnings_growth_12m_lag1`: 12-month earnings growth, lagged one period (annual interpolation; staleness flagged).
- `approvals_growth`: mortgage approvals 12-month growth.
- `affordability_gap_lag1`: price-to-income ratio deviation from long-run average, lagged one period.

4 Fair-Value OLS Model

4.1 Specification

The fair-value layer estimates equilibrium log real prices using a pooled OLS regression on anchor months:

$$\log P_{r,t}^{\text{real}} = \alpha + \beta_1 \log(\text{income}_{r,t}) + \beta_2 \log(\text{rent}_{r,t}) + \beta_3 \text{mortgage_rate}_{r,t} + \delta_r + \varepsilon_{r,t}, \quad (9)$$

where δ_r are regional fixed effects absorbed through regional dummy variables, and HAC (Newey-West, 6 lags) standard errors correct inference for autocorrelation.

Estimated coefficients ($n = 452$, $R^2 = 0.9496$):

$$\begin{aligned} \hat{\beta}_1 &= -1.1732 && (\log \text{income}), \\ \hat{\beta}_2 &= +1.7443 && (\log \text{rent}), \\ \hat{\beta}_3 &= -0.0104 && (\text{mortgage rate, level}). \end{aligned}$$

The estimated fair value in nominal terms is

$$\hat{P}_{r,t}^{\text{nom}} = \exp(\widehat{\log P_{r,t}^{\text{real}}}) \times \frac{\text{CPI}_t}{\text{CPI}_{\text{base}}}. \quad (10)$$

The log valuation gap is $x_{r,t} = \log P_{r,t}^{\text{real}} - \log \hat{P}_{r,t}^{\text{real}}$.

4.2 Gauss-Markov Theorem

Theorem 4.1 (Gauss-Markov). *Under the classical linear model assumptions—exogeneity ($\mathbb{E}[\varepsilon | X] = 0$), homoskedasticity ($\text{Var}(\varepsilon | X) = \sigma^2 I$), and no perfect multicollinearity—the OLS estimator $\hat{\beta} = (X^\top X)^{-1} X^\top y$ is the Best Linear Unbiased Estimator (BLUE).*

Proof. Unbiasedness.

$$\mathbb{E}[\hat{\beta}] = (X^\top X)^{-1} X^\top \mathbb{E}[y] = (X^\top X)^{-1} X^\top X \beta = \beta.$$

Minimum variance (BLUE). Let $\tilde{\beta} = Cy$ be any other linear unbiased estimator. Write $C = (X^\top X)^{-1} X^\top + D$ for some matrix D . Unbiasedness requires $\mathbb{E}[\tilde{\beta}] = \beta$, hence $DX = 0$. The variance of $\tilde{\beta}$ is

$$\text{Var}(\tilde{\beta}) = \sigma^2 CC^\top = \sigma^2 [(X^\top X)^{-1} + DD^\top].$$

Since $DD^\top$ is positive semi-definite, $\text{Var}(\tilde{\beta}) \succeq \sigma^2 (X^\top X)^{-1} = \text{Var}(\hat{\beta})$ for every linear unbiased estimator $\tilde{\beta}$. $\square$

HAC Correction and Gauss-Markov Violation

Housing price residuals violate the Gauss-Markov homoskedasticity assumption: all 12 regions exhibit significant autocorrelation and ARCH-type conditional heteroskedasticity (Section 7). HAC (Newey-West, 6 lags) standard errors restore asymptotically valid inference by adjusting $\text{Var}(\hat{\beta})$ for the observed autocorrelation and heteroskedasticity structure. Point estimates $\hat{\beta}$ remain consistent; only standard errors change. Residual autocorrelation in housing is well-documented [1] and is the primary motivation for HAC throughout.

4.3 PTI/PTR Blend Weight

The fair-value blend weight w_{PTI} allocates between a price-to-income (PTI) channel and a price-to-rent (PTR) channel. The weight is estimated by grid-search over $w \in [0.25, 0.75]$ minimising pooled squared log-price deviation:

$$\hat{w}_{\text{PTI}} = \arg \min_{w \in [0.25, 0.75]} \sum_{r,t} (\log P_{r,t} - w \log \text{income}_{r,t} - (1-w) \log \text{rent}_{r,t})^2. \quad (11)$$

The estimated value is $\hat{w}_{\text{PTI}} = 0.25$, clipped to the lower bound, indicating that the rental parity channel dominates in this panel—consistent with the large estimated $\hat{\beta}_2 = 1.744$ on log rent.

4.4 Mean-Reversion Speed Estimation

The mean-reversion speed κ_r is estimated from an AR(1) regression on the log valuation gap:

$$x_{r,t} = \phi_r x_{r,t-1} + \eta_{r,t}. \quad (12)$$

The annualised speed is $\kappa_r = -12 \log \hat{\phi}_r$. Regional estimates range from $\kappa = 0.131$ (West Midlands) to $\kappa = 0.825$ (Northern Ireland).

4.5 Volatility Estimation

Annual volatility σ_r is blended from raw and winsorised realised monthly volatility:

$$\sigma_r^{\text{blend}} = \text{clip}(0.35 \sigma_r^{\text{raw}} + 0.65 \sigma_r^{\text{win}}, 0.025, 0.12). \quad (13)$$

An IQR prior raises σ_r when the implied 5-year distribution is too narrow relative to historical 5-year return dispersion (threshold: IQR ratio ≥ 0.30). A sigma floor override is applied to South West ($\sigma \rightarrow 0.045$, raised from the fitted 0.030) to enforce the IQR ratio requirement.

4.6 Calibrated SDE Parameters by Region

Table 1: Region-level SDE parameters, bank-grade canonical rebuild (2026-04-02). κ = annualised mean-reversion speed; σ = annual log-price volatility; $t_{1/2} = \ln 2 / \kappa$ in years.

Region	κ	σ	$t_{1/2}$ (yr)	Note
East Midlands	0.419	0.080	1.65	
East of England	0.218	0.100	3.18	
London	0.200	0.051	3.47	rate ≈ 0 in OLS
North East	0.497	0.120	1.40	highest σ
North West	0.314	0.095	2.21	
Northern Ireland	0.825	0.115	0.84	fastest mean reversion
Scotland	0.468	0.044	1.48	
South East	0.227	0.085	3.05	
South West	0.458	0.030	1.51	floor $\rightarrow 0.045$ applied
Wales	0.214	0.040	3.24	
West Midlands	0.131	0.055	5.29	slowest mean reversion
Yorkshire and The Humber	0.326	0.050	2.13	

5 Stochastic Simulation

5.1 GBM vs Log-OU: Conceptual Comparison

Property	GBM	Log-OU
Mean reversion	No	Yes ($\kappa > 0$)
Long-run equilibrium	None	Fair-value path m_t
Log-price stationary distribution	None	$\mathcal{N}(m, \sigma^2 / (2\kappa))$
Positivity	Yes	Yes (log-price space)
Role in this model	Benchmark	Primary process

5.2 Target Mean Path Construction

The simulation mean path m_t is constructed as a linear blend between the current fair-value anchor P_0^* and the scenario terminal target P_T^* :

$$m_t = (1 - s_t) \log P_0^* + s_t \log P_T^*, \quad s_t = \frac{t}{T}. \quad (14)$$

The scenario terminal target is formed from the baseline five-year log return adjusted by macro overlays:

$$\log R_{\text{scen}}^{(5y)} = \log R_{\text{base}}^{(5y)} + 5 \times (3\hat{\beta}_{\text{rate}} \Delta r + 12\hat{\beta}_{\text{stress}} \Delta f + \Delta \gamma), \quad (15)$$

where Δr is the scenario rate shift (pp), Δf is the stress overlay, and $\Delta \gamma$ is the additional drift shift.

5.3 Five Macro Scenarios

Scenario	Rate shift (pp)	Drift shift (pp yr ⁻¹)
Baseline	0.0	0.0
Soft Landing	-0.5	+0.5
Rate Shock	+2.0	-1.5
Stagflation	+1.5	-2.0
Recovery Boom	-1.0	+2.0

5.4 Simulation Procedure

For each of the 12 regions and 5 scenarios, the Euler-Maruyama scheme (8) is applied with:

- $N = 10,000$ independent paths;
- $T = 60$ monthly steps ($\Delta t = 1/12$);
- i.i.d. standard normal draws ε_{n+1} from a fixed random seed for reproducibility.

Output statistics per region-scenario: median 5-year growth, $p_{10}/p_{25}/p_{75}/p_{90}$ percentiles, and the probability of a terminal loss exceeding 10%.

6 Scoring Engine

6.1 Signal Components

Let g_r denote the current percentage valuation gap (positive = overvalued relative to P^*). Define:

$$C1_{\text{mispricing}} = \text{clip}(50 - 0.95 \times g_r, 0, 100), \quad (16)$$

$$C2_{\text{consumer}} = \text{clip}(50 + 1.8 \times \text{WeightedConsumerReturn}, 0, 100), \quad (17)$$

$$C2_{\text{REIT}} = \text{clip}(50 + 1.8 \times \text{WeightedREITReturn}, 0, 100), \quad (18)$$

$$\text{YieldScore} = \text{clip}(50 + 16(\text{GrossYield} - 5.5), 0, 100), \quad (19)$$

$$\text{TailScore} = \text{clip}(100(1 - p_{\text{loss} > 10\%}), 0, 100). \quad (20)$$

6.2 Composite Scores

$$\text{ConsumerScore} = \text{clip}(0.40 C1_{\text{mispricing}} + 0.35 C2_{\text{consumer}} + 0.25 \text{TailScore}, 0, 100), \quad (21)$$

$$\text{REITScore} = \text{clip}(0.20 C1_{\text{mispricing}} + 0.35 C2_{\text{REIT}} + 0.25 \text{YieldScore} + 0.20 \text{TailScore}, 0, 100). \quad (22)$$

6.3 Scenario Aggregation Weights

Consumer aggregation is downside-skewed; REIT aggregation is equal-weighted:

$$w^{\text{consumer}} = (0.40, 0.25, 0.20, 0.10, 0.05) \quad (\text{Baseline, Rate Shock, Stagflation, Soft Landing, Boom}),$$

$$w^{\text{REIT}} = (0.20, 0.20, 0.20, 0.20, 0.20).$$

6.4 Score Labels and Governance

Score range	Consumer label	REIT label
≥ 62	Supportive	Supportive
50–61	Mixed	Mixed
< 50	Cautious	Cautious

Score Label Change — Risk Register R004 (RESOLVED)

Labels were changed from “Buy / Hold / Don’t Buy” to “Supportive / Mixed / Cautious” to reflect their nature as descriptive research buckets, not investment recommendations. The dashboard explicitly states that labels do not constitute investment advice. This change is documented in `SCORING_BACKTEST_NOTE.md` and risk register item R004, resolved 2026-04-02.

7 Validation and Backtesting

7.1 OLS Replay Deviation

The canonical rebuild re-estimates the fair-value OLS from the panel and compares fitted values against deployed outputs:

$$\max_{r,t} |\log \hat{P}_{r,t}^{\text{deployed}} - \log \hat{P}_{r,t}^{\text{replay}}| = 0.000$$

(to machine precision), confirming that all deployed fair-value outputs are exactly reproducible from source data and code.

7.2 Subsample OLS Validation

Training set: observations before 2018-01-01. Holdout: 2018-01-01 to 2026-01-01 ($n = 1,079$).

Model	RMSE (1m growth)	Dir. accuracy (12m)
OLS growth model	0.02165	64.7%
Zero-growth benchmark	0.01526	—

Short-Horizon RMSE Disadvantage

The OLS growth model does not beat the zero-growth benchmark at 1-month RMSE. This is expected and consistent with weak-form efficiency in liquid residential markets: monthly price changes are largely uninformative. The model's value-add is at horizons ≥ 24 months, where mean reversion dominates. Directional accuracy of 64.7% at the 12-month horizon exceeds the 55% governance threshold and is the primary evidence of model skill.

7.3 Sensitivity Analysis (One-At-a-Time Shocks)

Shocks applied to the fair-value OLS specification (9). Responses are uniform across all 12 regions because the OLS specification uses pooled coefficients.

Shock	Expected sign	Average ΔP^* (%)	Sign correct?
Income +10%	+	−10.58	NO
Income −10%	−	+13.16	NO
Rates +100 bp	−	−0.01	YES
Rates −100 bp	+	+0.01	YES
Rent +10%	+	+18.09	YES
Rent −10%	−	−16.79	YES

Income Sign Violation (Frisch-Waugh-Lovell Artefact)

The OLS income coefficient $\hat{\beta}_1 = -1.173$ is negative, producing sign violations on income shocks. By the Frisch-Waugh-Lovell theorem, the partial regression coefficient on log income in a specification that *already includes* log rent absorbs only the residual variation in income after partialling out rent. Since income and rent are positively correlated cross-sectionally, this residual carries negative partial correlation with prices in the panel. This is an econometric artefact of the semi-structural design, not a model defect. The income variable is retained for specification consistency with the broader literature. Rent (+18% per +10% rent shock) and rate channels produce economically correct signs and magnitudes.

7.4 Confidence Interval Coverage

A rolling 1-step-ahead test checks whether actual log prices fall within the model's 90% predictive interval:

$$|x_{t+1} - \mathbb{E}[x_{t+1} | x_t]| \leq 1.645 \sigma \sqrt{\Delta t},$$

where $\mathbb{E}[x_{t+1} | x_t] = x_t e^{-\kappa \Delta t} + \gamma \Delta t$ and the time-varying fair value m_{t+1} is taken from the in-sample OLS fair-value panel. Holdout: 2018-01-01 to 2026-01-01 (96 months per region).

Table 2: Rolling 1-step-ahead 90% CI coverage by region (holdout 2018–2026). Target = 0.90; pass threshold ≥ 0.80 ; material threshold = 0.70.

Region	Coverage	Assessment
East Midlands	0.854	Acceptable
East of England	0.865	Acceptable
London	0.740	Monitor
North East	0.854	Acceptable
North West	0.854	Acceptable
Northern Ireland	0.885	Acceptable
Scotland	0.698	Monitor
South East	0.875	Acceptable
South West	0.594	Below material threshold
Wales	0.677	Monitor
West Midlands	0.823	Acceptable
Yorkshire and The Humber	0.729	Monitor
Pooled (mean)	0.787	MARGINAL

CI Coverage Finding

Pooled coverage of 0.787 lies between the 0.70 material threshold and the 0.80 pass threshold. South West (0.594) and Scotland (0.698) are the weakest regions. Recalibrating σ to inflate coverage would constitute grade-shopping and is not recommended; the sigma floor override

for South West (raised to 0.045) addresses the root cause at the parameter level. Monitoring is recommended across all four flagged regions.

7.5 Residual Diagnostics

Fair-value OLS residuals are tested per region using: Ljung-Box (lags 1, 6, 12) for autocorrelation; Engle's ARCH LM test (lag 4) for conditional heteroskedasticity; and the Jarque-Bera test for non-normality. Statistics in bold have $p < 0.05$.

Table 3: Residual diagnostics (fair-value OLS, anchor months).

LB_k = Ljung-Box statistic at lag k ; ARCH = Engle ARCH-LM statistic (lag 4); JB = Jarque-Bera statistic.

Region	n	LB_1	LB_6	LB_{12}	ARCH	JB
East Midlands	41	31.9	119.9	142.7	34.3	0.27
East of England	41	35.6	143.4	182.6	33.9	2.48
London	41	33.6	134.9	170.1	34.4	0.84
North East	41	36.7	139.4	164.7	35.7	1.45
North West	41	35.3	142.7	171.9	32.5	1.31
Northern Ireland	21	14.8	35.0	42.0	16.4	1.85
Scotland	29	22.1	54.1	55.6	22.9	2.86
South East	41	33.7	130.0	164.4	35.0	1.40
South West	41	33.2	121.6	142.2	35.8	0.44
Wales	33	28.3	103.0	113.9	28.0	2.19
West Midlands	41	35.1	144.2	178.3	33.1	1.41
Yorkshire and The Humber	41	35.0	140.0	169.8	35.2	1.13

All 12 regions flag autocorrelation at all three lags and ARCH-type conditional heteroskedasticity. Zero regions fail the Jarque-Bera normality test. Residual autocorrelation in housing price series is well-documented [1] and is the primary motivation for HAC standard errors throughout. These diagnostic flags confirm that HAC correction is necessary and in place; they are not model defects.

7.6 Model Horse Race

Four models are compared on holdout 2018-01-01 to 2026-01-01 (pooled across 12 regions, 96 observations per region). GBM and random walk use 500 simulated paths; the median path is the point forecast. Log-OU uses the analytical conditional expectation.

Model	RMSE (1m)	RMSE (12m)	Dir. Acc. (1m)	Dir. Acc. (12m)
Log-OU	0.01445	0.04442	0.554	0.806
GBM	0.01449	0.04881	0.575	0.790
Zero-growth	0.01466	0.05582	0.055	0.000
Random walk	0.01472	0.05612	0.448	0.501

Key findings:

1. At the 12-month horizon, Log-OU achieves the lowest RMSE (0.04442 vs GBM 0.04881 and zero-growth 0.05582) and the highest directional accuracy (80.6% vs 79.0% for GBM).
2. At the 1-month horizon, zero-growth has the lowest RMSE, consistent with weak-form efficiency—expected and documented.
3. Log-OU outperforms all three benchmarks at the 12-month horizon, demonstrating that the OU mean-reversion structure adds measurable value at planning-relevant horizons.

8 Dashboard: UI Section Guide

The Streamlit research dashboard is structured as eight navigable pages.

8.1 Page 1 — Research Overview

Displays the model validation status, effective data date, and a summary scorecard across all 12 regions. Consumer and REIT scores with Supportive/Mixed/Cautious labels are shown in a regional summary table. An explicit disclosure banner states that model output is for research purposes only and does not constitute investment advice.

8.2 Page 2 — Regional Signals

Detailed view for a selected region. Shows: the current fair-value gap (x_r) and a price vs P^* chart; score decomposition with labelled C1/C2/Yield/Tail components; and evidence-level classification (Validated/Partial/Contextual) for each signal component.

8.3 Page 3 — Region Comparison

Side-by-side bar charts of consumer and REIT scores across all 12 regions. Supports ranking by score, valuation gap, or directional accuracy. Designed to support relative regional positioning for portfolio allocation.

8.4 Page 4 — Scenario and Diagnostics

Shows the five macro scenarios for a selected region: fan charts ($p_{10}/p_{25}/\text{median}/p_{75}/p_{90}$) of simulated price paths and terminal return distributions. Contains the interactive Scenario Lab, labelled explicitly as a “coefficient-based perturbation tool” and not a full econometric re-run.

8.5 Page 5 — Household Explorer

Interactive tool allowing a user to input a purchase price, deposit, and household income to see personalised affordability metrics and how current model signals apply to their budget range. Carries an explicit disclaimer that outputs are for illustrative research purposes only.

8.6 Page 6 — Methodology

Plain-language documentation of the seven-stage pipeline, fair-value OLS specification, simulation setup, and scoring weights. Points to this technical report for full mathematical derivations and governance details.

8.7 Page 7 — Technical Controls

Exposes the effective data date, earnings staleness months, model validation status, simulation seed, and per-region SDE parameters. Allows toggling between Baseline and alternative scenario views.

8.8 Page 8 — Limitations

Dedicated page listing all material limitations from `MODEL_LIMITATIONS.md`: regional-average vs. individual-property scope; interpolated annual series; omitted-variable risk; Scenario Lab limitations; and score compression. This page is prominently accessible from the main navigation and not hidden.

9 Model Governance

9.1 Sign-Off Status

As of 2026-04-02, all nine sign-off areas are approved by the model owner.

Area	Description	Status
G1	Data pipeline and panel integrity	APPROVED
G2	OLS fair-value specification	APPROVED
G3	SDE parameter estimation	APPROVED
G4	Monte Carlo simulation correctness	APPROVED
G5	Scoring engine calibration	APPROVED
G6	Validation and diagnostics pack	APPROVED
G7	UI disclosure and limitations	APPROVED
G8	Risk register completeness	APPROVED
G9	Reproducibility (replay deviation = 0)	APPROVED

9.2 Risk Register

The risk register contains 26 items: 2 RESOLVED, 2 MONITORED, 5 PARTIALLY_RESOLVED, 17 open/legacy.

Key resolved items as of 2026-04-02:

- **R002 (RESOLVED):** Legacy scripts archived to `archive/legacy_model/`. The `src/model/` directory now contains 7 canonical files only. No legacy imports remain in tests.
- **R004 (RESOLVED):** Score labels changed from Buy/Hold/Don't Buy to Supportive/-Mixed/Cautious; UI disclosure added; documented in `SCORING_BACKTEST_NOTE.md`.

9.3 Model Owner Review Schedule

Quarterly. Next review: 2026-06-30.

9.4 Deployment Restrictions

Deployment Restrictions

This model **may not** be used for: (i) individual property valuations; (ii) short-horizon trading decisions (horizon <24 months); (iii) regulatory capital calculations; or (iv) any purpose requiring a stress-testing model as defined under PRA SS3/18. These restrictions are disclosed in the UI Limitations page, in this report, and in the risk register.

10 Limitations and Future Work

10.1 Current Limitations

- L1. Regional averages only.** The model operates on Land Registry regional average prices. It is not applicable to individual properties, sub-regional markets, or property-type segments.

- L2. Interpolated annual series.** Earnings and net housing additions are available at annual frequency and are linearly interpolated to monthly. Within-year timing precision is therefore limited.
- L3. Omitted-variable and regime risk.** Despite HAC correction, omitted macro variables (credit conditions, planning policy shocks, stamp duty changes) and structural breaks remain genuine risks.
- L4. Scenario Lab is a perturbation tool.** The interactive Scenario Lab applies coefficient-based overlays to existing Stage 5 outputs; it is not a full econometric re-run with new data.
- L5. Score compression.** Composite scores reduce multi-dimensional distributional information to a single number. Evidence-level classifications (Validated/Partial/Contextual) partially offset this.
- L6. CI coverage marginal.** Pooled 1-step-ahead coverage of 78.7% is below the 80% pass threshold. South West and Scotland require monitoring.
- L7. Income sign violation.** The OLS income coefficient is negative (Frisch-Waugh-Lovell artefact); the model relies on the rent channel for income-affordability signal.
- L8. Earnings staleness.** At the modelling date, earnings data are 13 months stale across all regions.

10.2 Future Extensions

- F1.** Sub-regional and property-type granularity using ONS sub-regional HPI.
- F2.** Higher-frequency earnings proxies (e.g. PAYE real-time information) to reduce the 13-month staleness.
- F3.** Regime-switching extension replacing the fixed- κ OU process with a hidden Markov model for boom/bust regimes.
- F4.** Formal structural break test using the Andrews (1993) supremum- F procedure [4].
- F5.** Dynamic scenario weights incorporating macro-financial nowcasting signals rather than fixed judgement-based probabilities.

11 Conclusion

This report has documented the UK Regional Housing Market Model v0.2.0 from its probability-theoretic foundations through to production validation and governance. The principal achievements are:

1. A complete, reproducible seven-stage pipeline from raw Land Registry and ONS data to an interactive Streamlit research dashboard.
2. A semi-structural OLS fair-value layer ($n = 452$, $R^2 = 0.9496$) with HAC standard errors and documented sign-violation disclosure.
3. Region-level log-OU SDE parameters ($\kappa \in [0.131, 0.825]$, $\sigma \in [0.030, 0.120]$) estimated from AR(1) gap regressions and volatility blending.
4. A log-OU simulation framework that outperforms GBM and zero-growth at the 12-month horizon: RMSE 0.04442 vs 0.04881 (GBM) and 0.05582 (zero-growth), directional accuracy 80.6% vs 79.0% and 0.0%.
5. A bank-grade validation pack: replay deviation zero; subsample directional accuracy 64.7% at 12 months (above 55% threshold); CI coverage 78.7% (MARGINAL, monitored); 100% Ljung-Box and ARCH flagging mitigated by HAC standard errors.
6. A full governance framework: 9/9 sign-off areas approved; 26-item risk register; score label governance; explicit deployment restrictions.

The Euler-Maruyama log-OU process with a time-varying fair-value mean path is the central modelling choice that makes the simulation economically coherent and superior to GBM at planning-relevant horizons. The model is suitable for systematic institutional research into UK regional housing dynamics and long-horizon scenario analysis for REIT portfolio management.

References

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- [5] Vasiček, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2), 177–188.